

JeongHyeon Kim

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RESEARCH INTEREST

My research interests lie in **control-oriented dynamics modeling** and **whole-body control** for humanoid robots. I am interested in uncovering the physical relationships that govern robot behavior for a given task and formulating dynamics models around those relationships, so that robots can coordinate full-body motion robustly as contact conditions, payloads, terrain, and external interactions vary.

EDUCATION

- Korea University (GPA: 3.78/4.0, 4.25/4.5)** *Mar.2025 – Present*
M.S. in Mechanical Engineering Seoul, Republic of Korea
- Korea University (GPA: 3.48/4.0, 3.9/4.5)** *2019 – 2025*
B.S. in Mechanical Engineering Seoul, Republic of Korea

EXPERIENCE

- Human-Machine Systems Lab, Korea University** *Mar.2025 – Present*
Graduate Research Student (M.S.) Seoul, Republic of Korea
 - Whole-body Adaptation for Humanoid Standing Manipulation:** formulated the shared physical effect of upper-body motion, payload, and external interaction on whole-body balance as a **Balance Demand Descriptor (BDD)** based on centroidal dynamics, and incorporated it directly into locomotion control so the controller explicitly accounts for changing balance demands
- Human-Machine Systems Lab, Korea University** *Mar.2024 – Feb.2025*
Undergraduate Research Student Seoul, Republic of Korea
 - Visual Perception for Quadrupedal Object Interaction:** designed a dual-camera perception pipeline (global camera for workspace-level observation, local camera for precise near-field sensing) calibrated into a consistent coordinate frame, delivering task-appropriate visual information in real time despite viewpoint changes and occlusion during locomotion
 - Real-time Architecture for a Generative-AI-based Mobile Manipulator:** designed the real-time communication framework connecting simulation, an LLM, the user interface, and the physical robot; reorganized data flow around the robot's control cycle to resolve cross-module latency and meet real-time constraints

PROJECTS

- Quadruped Robot Obstacle Avoidance (CCP) — Team Leader** *Mar.2024 – Jan.2025*
Computer Vision, Autonomous Navigation
 - Developed obstacle detection and avoidance pipeline for DOGZILLA quadruped robot
 - Implemented monocular ORB-SLAM for depth estimation and UV-disparity for obstacle segmentation
- Ladder-Crossing Robot — Team Leader** *Mar.2023 – Sep.2023*
Mechanism Design and Control
 - Designed the ladder-crossing mechanism and implemented the Arduino-based control code
- Solar Tracking Parasol — Team Leader** *Sep.2023 – Nov.2023*
Mechanism Design and Control
 - Designed the link mechanism and implemented the sun-tracking control code

TECHNICAL SKILLS

Languages: Python, C/C++, MATLAB
Robotics: ROS2, Gazebo, URDF/XACRO
Tools: OpenCV, Arduino, Creo, NX, AutoCAD, SolidWorks, Git
OS: Linux (Ubuntu)

HONORS AND AWARDS

Fellowships and Scholarships

- Master's Research Encouragement Grant**, National Research Foundation of Korea *2025*
Awarded to selected research-track graduate students for research funding support

Awards

- President's Award**, Creative Challenger Program (CCP), Korea University *Jan.2025*
Awarded for outstanding performance in autonomous obstacle avoidance for a quadruped robot
- Bronze Award**, Creative Comprehensive Design Competition, Korea University *Sep.2023*
Awarded for the Solar Tracking Parasol mechanism design project
- Design Award**, Capstone Design Fair, Korea University *Nov.2023*